ATC3xxx Series User Manual

Warning

***** DO NOT USE APT UPGRADE or DIST-UPGRADE on ATC series system. *****

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It may overwrite some files (bootloader, kernel, DTB, etc.) with standard Nvidia packages.

Histories:

Version	Description
v1.1.3	1. Modified “ATC Series OTA Update”. (2026.03.04)
v1.1.2	1. Modified “Recovery image of ATC series system”. (2026.02.03) 2. Modified “ATC Series OTA Update”. (2026.02.10)
v1.1.1	1. Added “Update bootloader” instruction. (2025/11/4)
v1.1.0	1. Added “ATC Series OTA Update” instruction. (2025/10/29)
v1.0.10	1. Updated GMSL2 camera support list. (2025.10.03)
v1.0.9	1. Updated GMSL2 camera support list. (2025.08.26)
v1.0.8	1. Updated GMSL2 camera support list. (2025.07.22)
v1.0.7	1. Removed “switch root from NVMe” chapter. (2025.06.18) 2. Added “ATC products tools” instruction. (2025.06.18)
v1.0.6	1. Modified recovery image of ATC series system. (2024.07.02)
v1.0.5	1. Added ATC3520 product. (2023.08.25)
v1.0.4	1. Added ATC3540 product. (2023.05.16) 2. Corrected the errors in this document.
v1.0.3	1. Added ATC3750 product. (2023.01.07) 2. Updated SDK and AI Demo Guide.
v1.0.2	1. Modified AI Demo. (2022.07.29)
v1.0.1	1. Modified Switch root and added install SDK component. (2022.07.28)
v1.0.0	1. First released. (2021.11.03)

NEXCOM

Content

1. Introduction

Thank you for using the Nexcom ATC series Product. When you boot up the system first time, you need to do some settings, like selecting the time zone, setting a user account, etc. After finishing the setting, you have to install MUT package, then you can use all of the functions of ATC series Product.

Initialization

1.1. System Configuration

When you boot up ATC product first time, you will see these screens as follows:

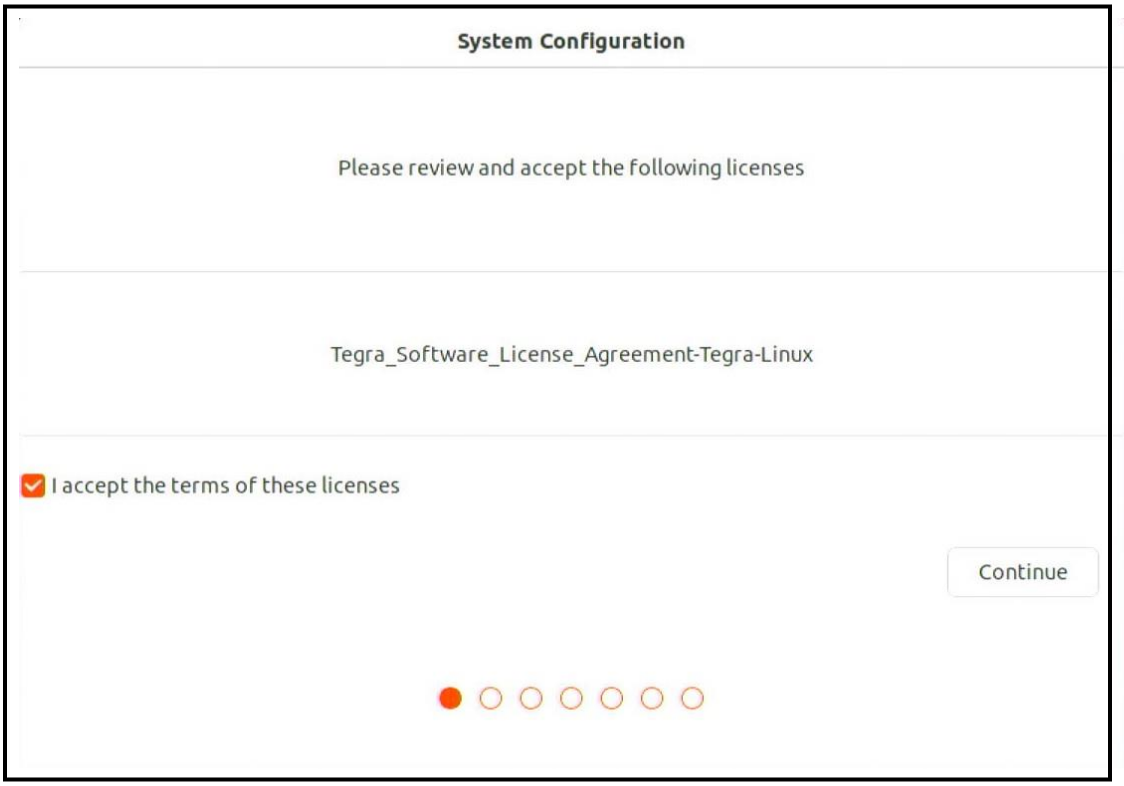


Figure 1. Check the terms.

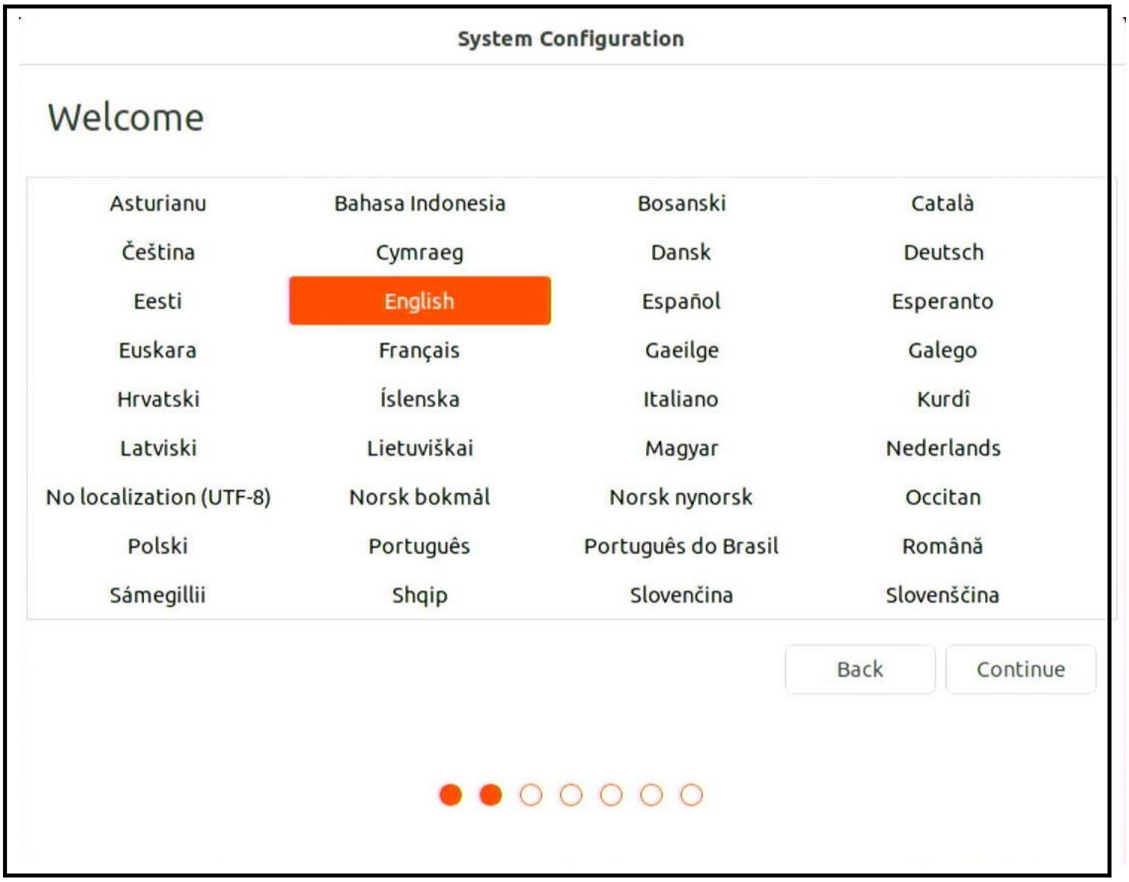


Figure 2. Select language

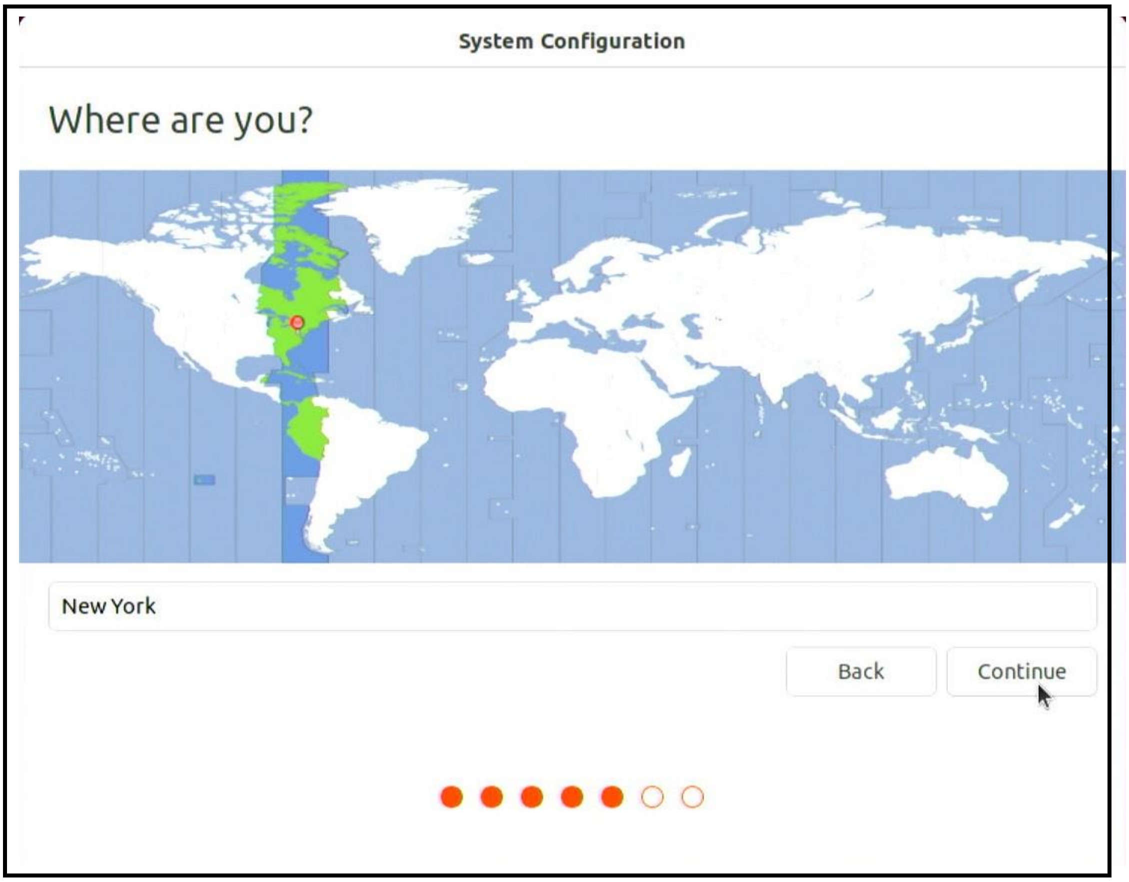


Figure 3. Select location

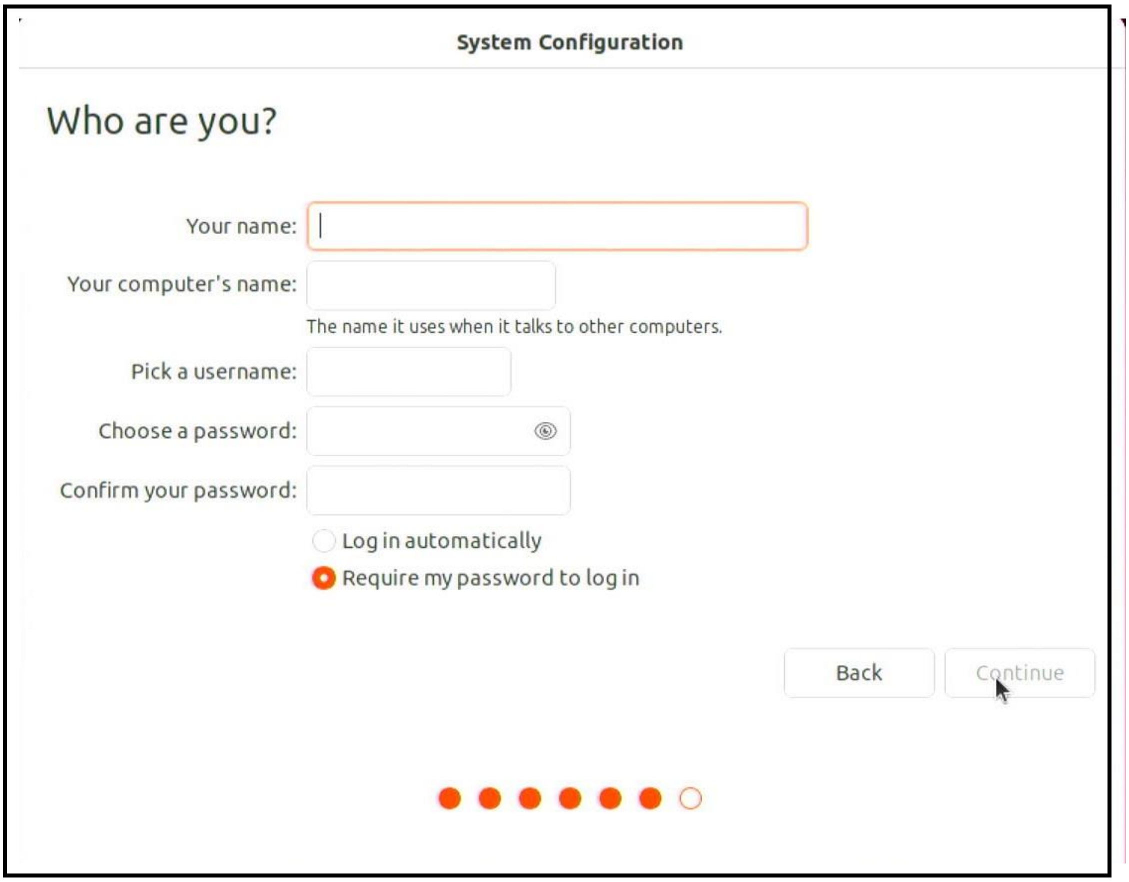


Figure 4. Enter name and password

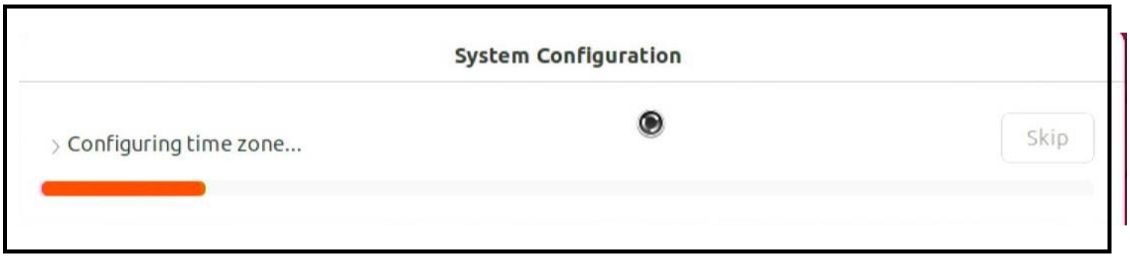


Figure 5. Download and update

After system configuration, you need install another necessary packages.

2. Recovery image of ATC series system

2.1. OS Firmware Flashing Integration overview

To address the issue of fragmented OS firmware packages for AGX Orin / Orin NX / Orin Nano platforms, we plan to consolidate the OS firmware into a single integrated package per SoM family.

OS Firmware Packages

SoM Family	OS Firmware Packages	Storage
AGX Orin	ATC375x_mfi_v4.x.x.x.tar.gz	eMMC/NVMe
AGX Orin Nano/NX	ATC356x_mfi_v4.x.x.x.tar.gz	NVMe

After integration, the OS firmware package supports the following features:

- Systems using the AGX Orin SoM can choose to flash the OS firmware to either eMMC or NVMe.
- Supports flashing systems with different carrier boards using the same OS firmware package. →example

⚠ MCU Firmware Requirement (Important)

- **Before using the integrated OS firmware, ensure MCU firmware version meets the minimum required version.**
- **If the MCU firmware version does not meet the requirement, an error will occur during MCU information verification, and the flashing process will not start.**

Carrier Board	MCU Version
ATC3750-6C / IP7-6C	AT375R15.bin
ATC3750-IP7-8M	AT376R04.bin

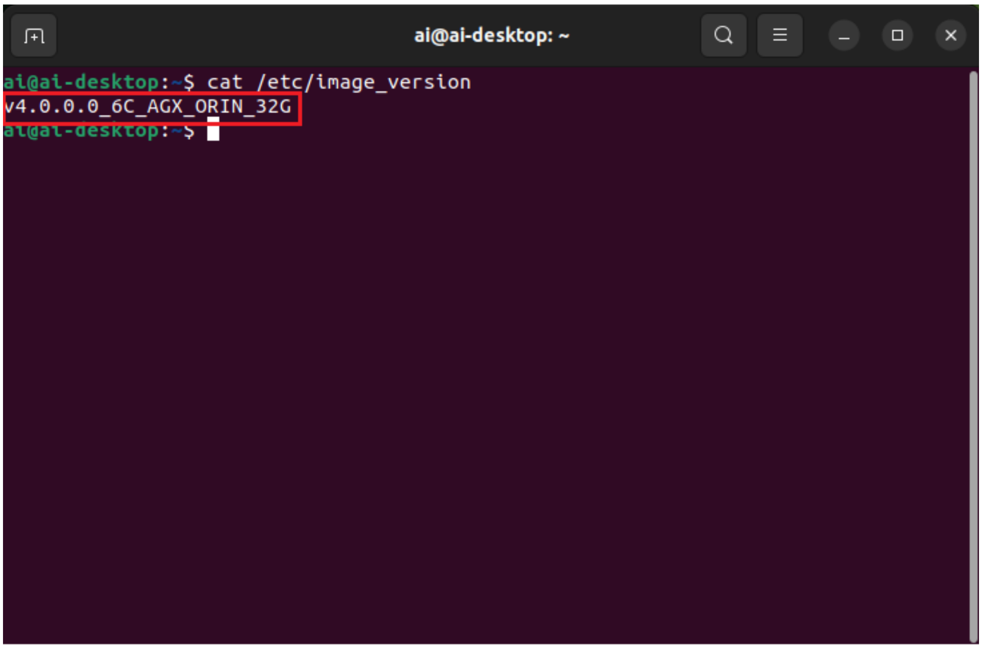
Carrier Board	MCU Version
ATC3560	AT353R31.bin
ATC3561	AT3561R01.bin
ATC3562	AT35620R02.bin
ATC3563	AT35630R01.bin
AIEdge-X80	AT353R26.bin

2.2. Prepare materials

1. A computer (Host) and install the **Ubuntu 18.04** or **20.04** system.
2. USB flash driver*1 (For images, the capacity depends on the size of the created image, and a minimum capacity of 3GB is required).
3. Micro USB cable.

2.3. Recovery

1. Boot up the ATC series system and check current image version.
   $ cat /etc/image_version



```
ai@ai-desktop:~$ cat /etc/image_version
v4.0.0.0_6C_AGX_ORIN_32G
ai@ai-desktop:~$
```

2. Find out the corresponding new version, e.g.
   ATC375x_mfi_v4.1.x.x.tar.gz / ATC356x_mfi_v4.2.x.x.tar.gz
3. In host PC, install sshpass package.
   $ sudo apt-get install sshpass

```
nexcom@Test-Server:~$ sudo apt-get install sshpass
[sudo] password for nexcom:
Reading package lists... Done
Building dependency tree
Reading state information... Done
sshpass is already the newest version (1.06-1).
The following package was automatically installed and is no longer required:
  shim
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 148 not upgraded.
nexcom@Test-Server:~$
```

4. Copy ATC375x_mfi_v4.1.11.0.tar.gz to host.

```
nexcom@Test-Server:~$ sudo cp /media/USB/ATC375x_mfi_v4.1.11.0.tar.gz /home/nexcom/
```

5. Untar the ATC375x_mfi_v4.1.11.0.tar.gz file.
   $ sudo tar xvf ATC375x_mfi_v4.1.11.0.tar.gz

Note. The commands need to be modified according to different file names.

⚠ DO NOT extract the firmware package using the "Archive Manager" application!!

```
nexcom@Test-Server:~$ sudo tar xvf ./ATC375x_mfi_v4.1.11.0.tar.gz
[sudo] password for nexcom:
./ATC375x_mfi_v4.1.11.0/
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin.conf
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/tsec_t234_sigheader.bin.encrypt
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/tegradevflash_v2
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/tegra234-bpmp-3701-0004-3737-0000_with_odm_sigheader.dtb.encrypt
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/tegrasign_v3_internal.py
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/uefi_jetson_with_dtb_aligned_blob_w_bin_sigheader.bin.encrypt
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/tegrabct_v2
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/mb1_bct_MB1_sigheader.bct.encrypt
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/tegrasign_v3_util.py
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/boot1.img
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/flashcmd.txt
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/nvpva_020_sigheader.fw.encrypt
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/boot4.img
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/applet_t234_sigheader.bin.encrypt
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/pyfdt/
./ATC375x_mfi_v4.1.11.0/atc3750-8M-orin/3701-0004/external/bootloader/pyfdt/__pycache__/
```

6. Using Micro USB cable connects host to ATC series system otg usb port.

7. Power on ATC series system and press the reset button immediately, wait for the LED on then release the reset button, after release reset button, the system into recovery mode.

8. Open the terminal in the host PC and type “lsusb” to check, if system has into recovery mode, you will see the information about nvidia.
   If not, please do the step 7 again.

```
nexcom@Test-Server:~$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 002: ID 0451:8440 Texas Instruments, Inc.
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 003: ID 0451:82ff Texas Instruments, Inc.
Bus 001 Device 010: ID 0955:7223 NVIDIA Corp.
Bus 001 Device 002: ID 0451:8442 Texas Instruments, Inc.
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

2.4. Start Flashing

1. Open the terminal and enter the unzipped folder of ATC3xxx_vx.x.x.x in host.

   $ cd ATC375x_mfi_v4.1.11.0

   Note. The commands need to be modified according to different file names.

2. Type commands to start flash.

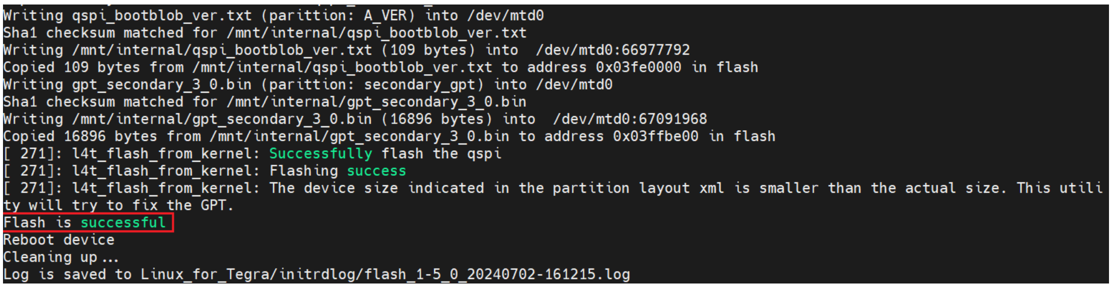
   For flashing AGX Orin to eMMC or Orin NX/Nano to NVMe：

   $ sudo ./tools/kernel_flash/l4t_initrd_flash.sh --flash-only --massflash 1

   For flashing AGX Orin to NVMe：

   $ sudo ./nvme.sh 1

3. In host pc, it will keep flashing until log show success.



```
Writing qspi_bootblob_ver.txt (parittion: A_VER) into /dev/mtd0
Sha1 checksum matched for /mnt/internal/qspi_bootblob_ver.txt
Writing /mnt/internal/qspi_bootblob_ver.txt (109 bytes) into  /dev/mtd0:66977792
Copied 109 bytes from /mnt/internal/qspi_bootblob_ver.txt to address 0x03fe0000 in flash
Writing gpt_secondary_3_0.bin (parittion: secondary_gpt) into /dev/mtd0
Sha1 checksum matched for /mnt/internal/gpt_secondary_3_0.bin
Writing /mnt/internal/gpt_secondary_3_0.bin (16896 bytes) into  /dev/mtd0:67091968
Copied 16896 bytes from /mnt/internal/gpt_secondary_3_0.bin to address 0x03ffbe00 in flash
[ 271]: l4t_flash_from_kernel: Successfully flash the qspi
[ 271]: l4t_flash_from_kernel: Flashing success
[ 271]: l4t_flash_from_kernel: The device size indicated in the partition layout xml is smaller than the actual size. This utili
ty will try to fix the GPT.
Flash is successful
Reboot device
Cleaning up ...
Log is saved to Linux_for_Tegra/initrdlog/flash_1-5_0_20240702-161215.log
```

4. Configure finished, the system will reboot into OS and [start system configuration.]

2.5. Multi Carrier Board Mass Flash Supports

The integrated OS firmware package supports mass flashing across different carrier boards, e.g.

- **Mass Flash OS Firmware to eMMC (AGX Orin SoMs)**

  If you have the following devices and want to flash the OS firmware to eMMC

  ATC3750-IP7-6C (32 GB) **× 2** units, ATC3750-IP7-8M (64 GB) **× 3** units

  Ensure that all systems have entered recovery mode, and modify the command in Step 2-4 as follows:

  $ sudo ./tools/kernel_flash/l4t_initrd_flash.sh --flash-only --massflash **5**

- **Mass Flash OS Firmware to NVMe (AGX Orin SoMs)**

  If you have the following devices and want to flash the OS firmware to NVMe

  ATC3750-IP7-6C (32 GB) **× 2** units, ATC3750-IP7-8M (64 GB) **× 3** units

  Ensure that all systems have entered recovery mode, and modify the command in Step 2-4 as follows:

  $ sudo ./nvme.sh **5**

- **Mass Flash OS Firmware to NVMe (Orin Nano / Orin NX SoMs)**
  If you have the following devices and want to flash the OS firmware to NVMe
  ATC3560 × 1 unit, ATC3561 **× 1** unit, ATC3562 **× 1** unit, ATC3563 **× 2** units
  Ensure that all systems have entered recovery mode, and modify the command in Step 2-4 as follows:
  $ sudo ./tools/kernel_flash/l4t_initrd_flash.sh --flash-only --massflash **5**

Note. The --massflash parameter specifies the maximum number of devices that can be flashed concurrently. (minimum = 1, maximum = 5)

⚠ When performing mass flashing on AGX Orin SoMs, all devices must be flashed to the same storage type (either eMMC or NVMe).

⚠ AGX Orin and Orin NX / Orin Nano devices can't be flashed at the same time. If the following message appears, it indicates that AGX Orin and Orin NX/Nano devices are in recovery mode simultaneously, which is not supported.

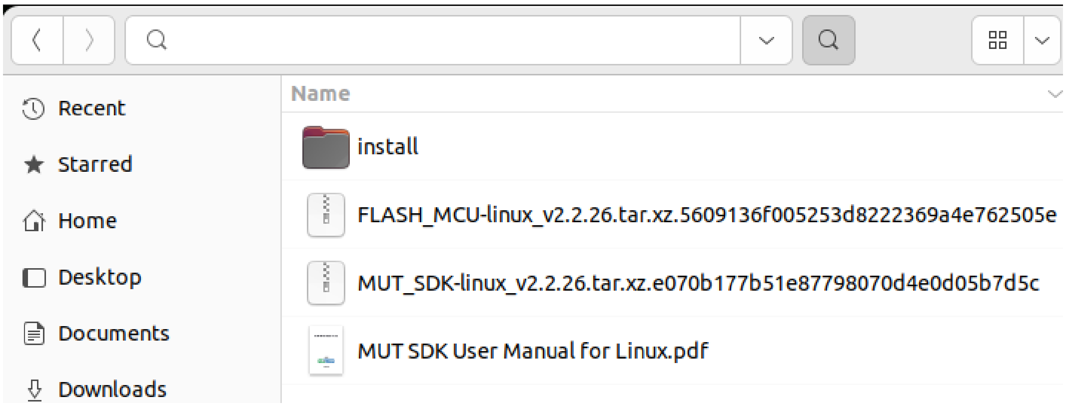
```
Error, there are different SoMs in RCM mode.
AGX Orin : 1, Orin NX/Nano : 1
```

3. ATC products tools

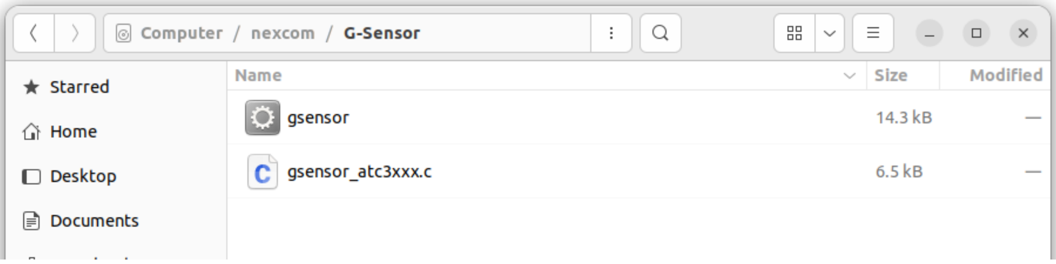
Open the terminal, there are installation packages and documents in the path as following: /nexcom/

3.1. MUT SDK

MUT SDK is a service to control the MCU of ATC series products, you can refer to the MUT SDK User Manuel document “ MUT_SDK_User_Manual_for_Linux.pdf ” in the MUT folder.



3.2. G-Sensor



The ATC series use ST LSM6DSL, which supports a 3D digital accelerometer and a 3D digital gyroscope. “gsensor_atc3xxx.c” is a sample code that can read accelerometer and gyroscope data.

Product	I2C-BUS	Addr
ATC35xx	i2c-7	0x6b
ATC37xx	i2c-0	0x6b

Usage : $ sudo -s

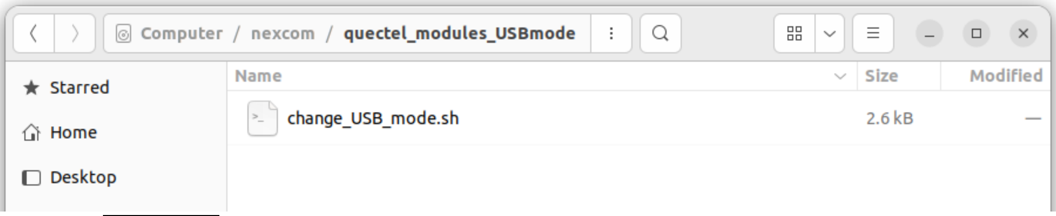
$ ./gsensor <save_output_as_file> <i2c-bus(only bus number)>

$ ./gsensor output.log 0

```
root@ai-desktop:/nexcom/G-Sensor# ./gsensor ./output.log 0
/dev/i2c-0
Initalize LSM6DSL successfully
ACC X:-0.01G     Y:-0.01G        Z:1.02G
GYRO X:94        Y:FE99  Z:FFEF
Left the machine for 3 seconds ...
ACC X:-0.02G     Y:-0.01G        Z:1.03G
GYRO X:95        Y:FE96  Z:FFED
Shut down LSM6DSL successfully
```

3.3. quectel_modules_USBmode

The ATC series supports several LTE/5G modules. This script is intended for Quectel modules(e.g. RM520N-GL, EM05-G) that require USB mode switching when running on linux OS.

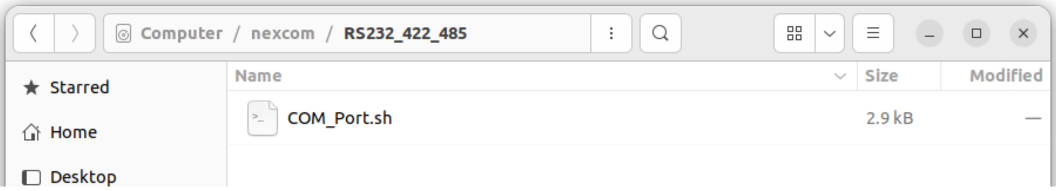


Usage : `$ sudo -s`

`$ bash ./change_USB_mode.sh 0`

3.4. RS232_422_485

This script only can run on the **ATC3750-8M**. COM1 supports RS232/422/485 conversion.



Usage : `$ sudo -s`

RS232 : `$ bash ./COM_Port.sh RS232`

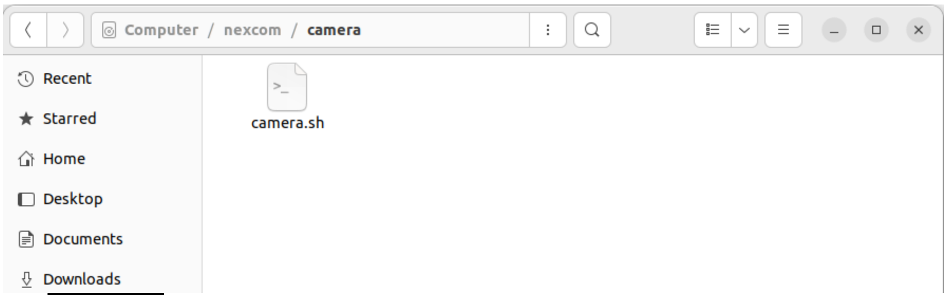
RS422 : `$ bash ./COM_Port.sh RS422`

RS485 – receive : `$ bash ./COM_Port.sh RS485-R`

RS485 – transmit : `$ bash ./COM_Port.sh RS485-W`

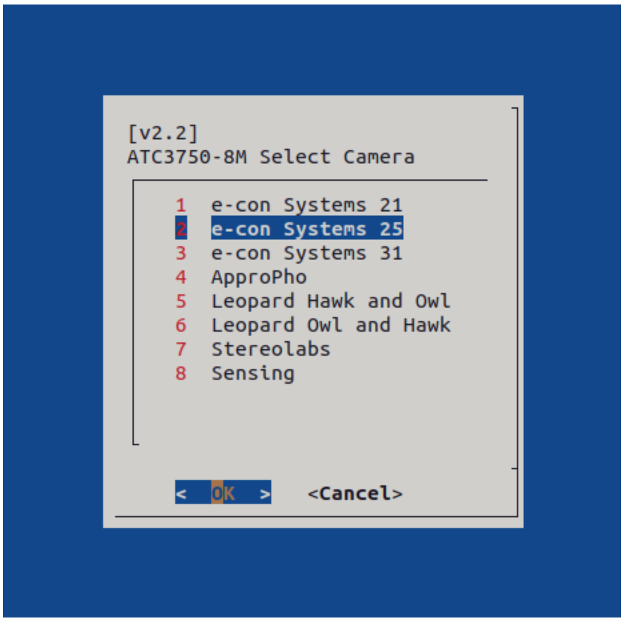
3.5. GMSL2_camera

<table>
<tr><th>GMSL2 Script Version</th><th colspan="5">GMSL2 Camera Support List</th></tr>
<tr><td>v1.0</td><td rowspan="5">e-con Systems
STURDeCAM21
STURDeCAM25
STURDeCAM31</td><td></td><td></td><td></td><td></td></tr>
<tr><td>v1.1</td><td rowspan="4">AppoPho
AP-AR0234
AP-IMX335
AP-IMX415</td><td></td><td></td><td></td></tr>
<tr><td>v2.0</td><td rowspan="3">Leopard Imaging
LI-AR0234-OWL
LI-AR0234-HAWK</td><td></td><td></td></tr>
<tr><td>v2.1</td><td rowspan="2">Stereolabs
ZED X</td><td></td></tr>
<tr><td>v2.2</td><td>Sensing
ISX031C-GMSL2F</td></tr>
</table>



Usage : $ sudo -s

$ bash ./camera.sh



After the driver and dts have been successfully updated, you need to power off the system and install the GMSL2 camera on CAM1~CAM8.

4. Install SDK component

Location: /nexcom/nvidia_SDK_components/install_sdk.sh Free disk space required: 10.6 GB (SDK 9924MB, Deepstream 733MB). The script will install SDK automatically following:

1. TensorRT
2. cuDNN
3. CUDA
4. Multimedia API
5. Computer Vision
6. Developer Tools (Nvidia Nsight System and Graphics)
7. Deepstream

```
$ sudo ./install_sdk.sh
```

4.1. How to check the SDK version

```
$ sudo apt install python3-pip
$ sudo pip3 install jetson_stats
$ sudo jtop(Figure 8.)
```

Move the cursor to “6INFO” and you can see the library version.

You can also use another command: (Figure 9)

```
$ jetson_release
```

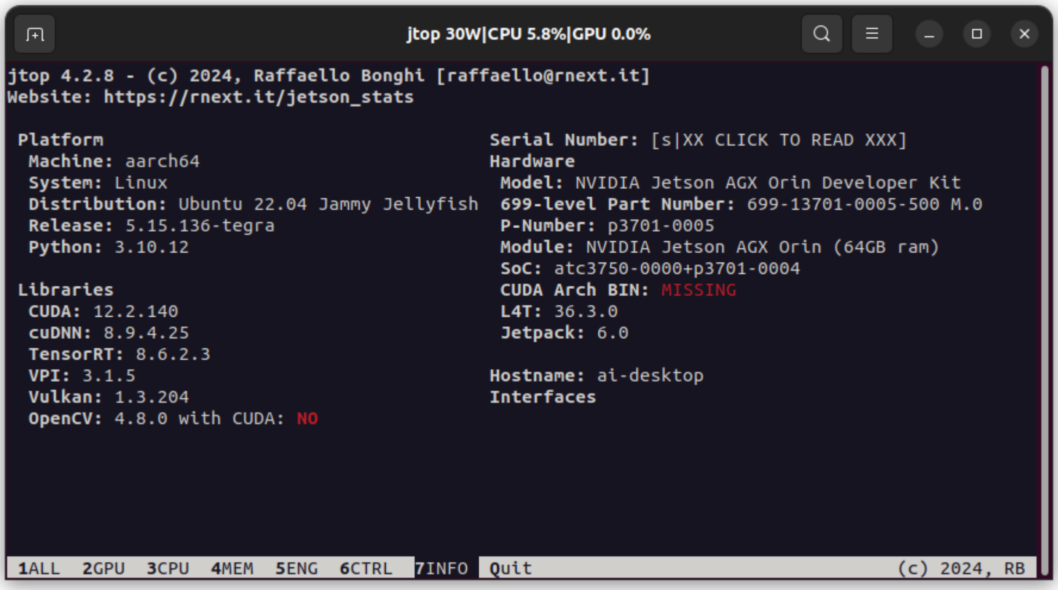


Figure 8. Info tab on jtop

```
root@ai-desktop:/home/ai# jetson_release
Software part of jetson-stats 4.2.8 - (c) 2024, Raffaello Bonghi
Model: NVIDIA Jetson AGX Orin Developer Kit - Jetpack 6.0 [L4T 36.3.0]
NV Power Mode[2]: MODE_30W
Serial Number: [XXX Show with: jetson_release -s XXX]
Hardware:
 - P-Number: p3701-0005
 - Module: NVIDIA Jetson AGX Orin (64GB ram)
Platform:
 - Distribution: Ubuntu 22.04 Jammy Jellyfish
 - Release: 5.15.136-tegra
jtop:
 - Version: 4.2.8
 - Service: Active
Libraries:
 - CUDA: 12.2.140
 - cuDNN: 8.9.4.25
 - TensorRT: 8.6.2.3
 - VPI: 3.1.5
 - Vulkan: 1.3.204
 - OpenCV: 4.8.0 - with CUDA: NO
root@ai-desktop:/home/ai#
```

Figure 9. Info of jetson_release

5. AI Demo

5.1. Installation

Use the install_sdk.sh (Chapter 5) to install deepstream and other SDK.

5.2. Run the deepstream demo

Run the demo as following command:

```
$ cd /opt/nvidia/deepstream/deepstream/samples/configs/deepstream-app
$ deepstream-app –c
source4_1080p_dec_inferresnet_tracker_sgie_tiled_display_int8.txt
```

The result

Figure 10. deepstream-app demo

6. ATC Series OTA Update

OTA payload package name	rootfs update block	nexcom OS version	Support status
ATC375x_v4.1.11.0 ota_payload_package.tar.gz	eMMC/NVMe	v4.x.x.x -> v4.1.11.0	✔
ATC356x_v4.2.5.0_ ota_payload_package.tar.gz	NVMe	v4.x.x.x -> v4.2.5.0	✔

* Nexcom OS **v4.1.x.x** corresponds to **JetPack 6.1** BSP, **v4.2.x.x** corresponds to **JetPack 6.2** BSP.

6.1. MCU Firmware Requirement(Important)

- **Before using the OTA update, ensure MCU firmware version meets the minimum required version.**
- **If the MCU firmware version does not meet the requirement, an error will occur during MCU information verification, and the OTA update will not trigger.**

Carrier Board	MCU Version
ATC3750-6C / IP7-6C	AT375R15.bin
ATC3750-IP7-8M	AT376R04.bin

Carrier Board	MCU Version
ATC3560	AT353R31.bin
ATC3561	AT3561R01.bin
ATC3562	AT35620R02.bin
ATC3563	AT35630R01.bin
AIEdge-X80	AT353R26.bin

6.2. Preparing the OTA payload package

- Contact the Nexcom FAE Team to obtain the files "<atc_rel>_ota_payload_package.tar.gz" and "ota_tools_<rel>_aarch64.tbz2".

⚠ **Do not download the ota_tools_<rel>_aarch64.tbz2 from nvidia jetson linux, as this may result in an incomplete update!!**

- Set the <WORKDIR> environment variable to the absolute path of your working directory.

```
$ mkdir -p /home/${USER}/work_dir
$ export WORKDIR=/home/${USER}/work_dir
```

- Unpack "ota_tools_<rel>_aarch64.tbz2" into ${WORKDIR}

```
$ sudo tar -C $WORKDIR -xvf ota_tools_<rel>_aarch64.tbz2
```

- Create /ota/ directory and place the <atc_rel>_ota_payload_package.tar.gz OTA payload package inside it.

```
$ sudo mkdir -p /ota/
$ sudo mv /home/${USER}/<atc_rel>_ota_payload_package.tar.gz /ota/
```

6.3. Start the OTA update

- Trigger the OTA update

```
$ cd ${WORKDIR}/Linux_for_Tegra/tools/ota_tools/version_upgrade
$ sudo ./nv_ota_start.sh /ota/<atc_rel>_ota_payload_package.tar.gz
```

⚠ **DO NOT POWER OFF the device or interrupt the process during the update!!**

⚠ **Any unexpected shutdown can cause update failure and data loss!!**

6.4. Update bootloader (For Orin-NX & Orin-Nano)

- If you are using Orin-NX or Orin-Nano and want to update the UEFI, please execute the following command:

```
sudo nv_bootloader_capsule_updater.sh -q /<ota_payload>/TEGRA_BL.Cap
```

For AGX Orin, the bootloader update is already included in nv_ota_start.sh, so there's no need to execute nv_bootloader_capsule_updater.sh.

6.5. Backing up and restoring files

- Before reboot, edit the configuration file ota_backup_files_list.txt (located at /ota_work/) to add the paths of files/directories that must be preserved after the OTA update.

NVIDIA's OTA update overwrites the rootfs in place. If you have data that needs to be kept, be sure to add those paths to avoid data loss. We have pre-populated a subset of essential items [Figure 11.]
(e.g., user configuration files, power-mode settings, GMSL2 camera DTBOs, etc.).

```
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# THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
# LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING
# FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER
# DEALINGS IN THE SOFTWARE.

# This file contains a list of files/directories in the APP partition
# that are to be backed up and restored during OTA.

# This file is used by the script "nv_ota_preserve_data.sh"

# All the files or directories should be listed with absolute path
# Example:
# etc/passwd
# opt/nvidia
etc/passwd
etc/shadow
etc/group
etc/gshadow
home/
etc/hostname
etc/hosts
etc/default/locale
etc/timezone
etc/sudoers
etc/sudoers.d/
etc/systemd/resolved.conf
etc/NetworkManager/system-connections/
var/lib/AccountsService/users/
var/lib/nvpmodel/
nexcom/camera/nexcom_cam_setting
```

Figure 11. ota_backup_files_list.txt

6.6. Reboot system

- After trigger the OTA update, reboot the system to begin the system update.

```
Verifying image /tmp/recovery-dtb_alt.tmp with sha1 chksum file /tmp/sha1sum.tmp
Sha1 checksum for /tmp/recovery-dtb_alt.tmp (456746166ddff86a786cb16a16f0d7acc57d5829) matches
Write /ota_work/internal_device/images-R36-ToT/tegra234-p3701-0004-atc3750-8M-base_ver.dtb.rec to recovery-dtb_alt(/dev/mmcblk0p12) succe
ssfully.
Start swapping partition name and guid.
Set name recovery-dtb_alt to /dev/mmcblk0p9 and recovery-dtb to /dev/mmcblk0p12.
Swap partition name successfully.
Start erasing recovery-dtb_alt.
Erase recovery-dtb_alt(/dev/mmcblk0p9) successfully.
Done. Update recovery-dtb successfully.
Updating recovery-dtb and recovery-dtb_alt partitions done
install_partition_with_alt /ota_work/internal_device/images-R36-ToT esp
prerequisite_check esp
The /ota_work/internal_device/images-R36-ToT/esp.img for partition esp is not found
Skip updating esp partition as no valid image is found
clean_up_boot_partition
update_rootfs /ota_work
update_rootfs_with_a_b_disabled /ota_work
update_rootfs_in_recovery /ota_work
force_booting_to_recovery
Force booting to recovery by writing \x07\x00\x00\x00\x03\x00\x00\x00 to UEFI variable L4TDefaultBootMode-781e084c-a330-417c-b678-38e6963
80cb9
dd if=/tmp/var_tmp.bin of=L4TDefaultBootMode-781e084c-a330-417c-b678-38e696380cb9 bs=8
1+0 records in
1+0 records out
8 bytes copied, 0.00228674 s, 3.5 kB/s
Rootfs is to be updated in recovery kernel once device is rebooted.
check_bootloader_version /ota_work
update_bootloader /ota_work
Bootloader on non-current slot(B) is to be updated once device is rebooted
clean_up_ota_files
ai@ai-desktop:~/version_upgrade$
```

Figure 12. reboot the system to begin the system update

```
$ sudo reboot -h now
```

6.7. Updating bootloader and rootfs

- If you want to observe the update status, you can connect the serial console port to host PC. **There is no console port on the ATC3750-IP7-8M.**

⚠ **DO NOT POWER OFF the device or interrupt the process during the update!!**

⚠ **Any unexpected shutdown can cause update failure and data loss!!**

```
Jetson System firmware version 202405.0-ca7f3d9f-dirty date 2025-10-22T06:20:44+
00:00
ESC   to enter Setup.
F11   to enter Boot Manager Menu.
Enter to continue boot.

Update Progress - 100% **************************************************▒▒▒▒Shutdown state requested 1
Rebooting system ...
```

Figure 13. Update progress done

6.8. Check OTA update result

- All update logs are saved under /last_ota_update_log directory. You can check these logs to confirm that the update was successful.
  You can also verify the result by checking /etc/image_version and confirming that it matches the version in <atc_rel>_ota_payload_package.tar.gz

Reference : OTA Update NVIDIA online documentation

https://docs.nvidia.com/jetson/archives/r36.4.3/DeveloperGuide/SD/SoftwarePackagesAndTheUpdateMechanism.html#steps-performed-on-the-jetson-device